

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May - June 2023

ME1003-1- ELEMENTS OF MECHANICAL ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note:

*Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.*

*Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. _____ is not a conventional source of energy.
 A) Coal
 B) Petrol
 C) Flowing water
 D) Solar
2. Pressure of the steam can be raised quickly in
 A) water tube boiler
 B) fire tube boiler
 C) in both water and fire tube boilers
 D) none of these
3. T-tube is used in Babcock and Wilcox boiler to
 A) Supply the steam to the super heater
 B) Supply steam to different applications
 C) Release the excess steam
 D) Supply water
4. The enthalpy of wet steam at dryness fraction of 0.98 is _____
 ($h_f = 762.61 \text{ kJ/kg}$, $h_{fg} = 2013.6 \text{ kJ/kg}$)
 A) 2013.6 kJ/kg
 B) 2776.5 kJ/kg
 C) 2522.2 kJ/kg
 D) 2735.9 kJ/kg
5. The steam is heated above its dry saturated state, then its temperature
 A) Decreases
 B) Increases
 C) Remains the same
 D) None of these
6. The strainer in centrifugal pump is for
 A) Priming
 B) Restricting impurities from entering impellar
 C) To decrease pressure
 D) None of these
7. Which of the following is not a part of open cycle gas turbine?
 A) Combustion chamber
 B) Heat exchanger
 C) Compressor
 D) Cooler
8. In a four stroke engine, the working cycle is completed in _____
 A) One revolution of the crankshaft
 B) Two revolutions of the crankshaft
 C) Three revolutions of the crankshaft
 D) Four revolutions of the crankshaft
9. A two stroke cycle engine gives _____ the number of power strokes as compared to the four stroke cycle engine, at the same engine speed.
 A) Half
 B) Same
 C) Double
 D) Four times
10. Creep is the phenomena due to
 A) Elasticity of the belt
 B) Slip between surfaces
 C) High tension in the belt
 D) Low tension in the belt
11. The teeth on the spur gears with respect to axes of gears are
 A) Parallel
 B) Perpendicular
 C) Inclined
 D) Intersecting
12. Heat is rejected by refrigerant during a refrigeration cycle in a
 A) Compressor
 B) Condenser
 C) Evaporator
 D) Expansion valve
13. Which of the following is not a desirable property of the refrigerant
 A) High boiling point
 B) Odourless
 C) Non-corrosive
 D) Non-toxic
14. The intermediate cone is observed in
 A) Reducing flame
 B) Neutral flame
 C) Oxidizing flame
 D) All of these

15. In lathe work, when the tool is fed parallel to the rotation of job work, it will produce
 A) Cylindrical surface B) Spherical surface
 C) Tapered surface D) All of these
16. Lathe machine produces ' _____ '
 A) Spherical surface B) Cylindrical surface
 C) Flat surface D) Both B and C
17. Automation is the application of technology
 A) To monitor the process B) To control the process
 C) To monitor and control the process D) To monitor and control the process with minimum human effort
18. The automated process in which the equipment configuration determines the sequence of processing (or assembly) actions.
 A) Fixed Automation B) Programmable Automation
 C) Flexible Automation D) Integrated Automation
19. A robot is a
 A) A general purpose machine B) A re-programmable machine
 C) A reprogrammable general purpose machine D) A customized machine used for a specific task
20. What form of automation is considered robotics?
 A) Fixed Automation B) Programmable Automation
 C) Flexible automation D) Programmable and flexible automation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	List four important boiler accessories and explain them in the relevant applications.	8	L1	1	1
b)	Explain the working of Kaplan turbine with neat diagram.	8	L2	2	1
2. a)	5 kg of wet steam of dryness 0.8 passes from a boiler to a super heater at a constant pressure of 1 MPa abs. In the super heater its temperature increases to 350°C. Determine the amount of heat supplied in the super heater. The specific heat of the super-heated steam $C_{PS} = 2.25 \text{ kJ/kg K}$. Take $T_S = 179.88^\circ\text{C}$, $h_f = 762.61 \text{ kJ/Kg}$ and $h_{fg} = 2013.6 \text{ kJ/Kg}$	4	L3	1	2
b)	Define renewable and non-renewable sources of energy with suitable examples.	4	L1	1	1
c)	Explain the working of single stage air-compressor with neat diagram.	4	L2	2	1
d)	List out the differences of impulse and reaction steam turbines.	4	L1	2	1
3. a)	Explain the constructional features and working of a water tube boiler.	8	L1	1	1
b)	Illustrate working principle and functioning of a centrifugal pump.	8	L1	2	1
Unit – II					
4. a)	With neat sketches describe the principle and working of a 4-stroke spark ignition engine.	8	L2	3	1
b)	Write a short note on following: i) Spur gear ii) Bevel gear iii) Helical gear iv) Worm gear	8	L1	4	1
5. a)	The following observations were obtained during a trial on a four-stroke diesel engine. Cylinder diameter = 25 cm, Stroke of the piston = 40 cm Crankshaft speed = 250 rpm Brake load = 70 kg Break drum diameter = 2 m Mean effective pressure = 6 bar Diesel oil consumption = 0.1 m ³ /min Specific gravity of diesel = 0.78 Find: (i) Brake Power, (ii) Indicated Power (iii) Friction Power (iv) Mechanical Efficiency (v) Brake thermal Efficiency (vi) Indicated Thermal Efficiency	8	L3	3	2
b)	List the properties of the good refrigerant.	8	L1	4	1

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|-------|---|---|----|---|---|
| 6. a) | In a crossed belt drive the difference in tensions between the tight and slack sides of the belt is 1000 N. Find the tension on the slack and tight sides, if the angle of contact is 160° and its coefficient of friction is 0.3. | 8 | L3 | 3 | 2 |
| b) | Define welding and compare the principle and features of welding and brazing. | 8 | L3 | 4 | 2 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Define automation and briefly describe the characteristics of Fixed, Programmable, and Flexible automation systems. | 8 | L2 | 5 | 1 |
| b) | With a neat sketch explain the following machining operations | | | | |
| i) | Turning | | | | |
| ii) | Taper turning | 8 | L2 | 5 | 1 |
| 8. a) | Describe the following machining operations with neat diagram. | | | | |
| i) | Milling | | | | |
| ii) | Thread cutting | 8 | L2 | 5 | 1 |
| b) | What is robotics? list few important applications of robots in the industry. | 4 | L2 | 5 | 1 |
| c) | Write a short note on additive manufacturing. | 4 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
